

Vladimir Strokov

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EDUCATION

Johns Hopkins University Department of Physics and Astronomy Ph.D. Advisor: Emanuele Berti / Co-advisor: Giacomo Fragione (Northwestern) Research: Gravitational Waves, Intermediate-Mass Black Holes, Populations of GW Sources	Baltimore, USA 2020–2024
Lebedev Physical Institute Astro Space Centre, Theoretical Astrophysics Department Candidate of Sciences , Advisor: V.N. Lukash Research: Gravity Theory	Moscow, Russia 2011
Moscow Institute of Physics and Technology Ginzburg's Chair for Problems of Physics and Astrophysics M.Sc. , <i>with excellence</i> / B.Sc. , <i>with excellence</i>	Moscow, Russia 2007 / 2005

RESEARCH

West Virginia University Center for Gravitational Waves and Cosmology (GWAC) GWAC Postdoctoral Fellow , LISA GW science	Morgantown, USA 2024–present
Lebedev Physical Institute Astro Space Centre, Theoretical Astrophysics Department Lab Assistant / Research Fellow (on medical leave: 2014–2017) Activities: - GPU algorithms for simulations of black hole shadows - Extraction of black hole silhouettes from future interferometric data - Selecting exoplanetary candidates for the Millimetron project	Moscow, Russia 2005–2020
Federal Univeristy of Espírito Santo J.C. Fabris' Gravitation and Cosmology Group Postdoctorate , theories of modified gravity	Vitória, ES, Brazil 2013–2014
Federal Univeristy of Juiz de Fora I.L. Shapiro's Field Theory Group Postdoctorate , field theory in curved space-time	Juiz de Fora, MG, Brazil 2012–2013

PROGRAMMING PROFICIENCY

The scientific research toolkit:

- Python: `numpy`, `matplotlib`, `astropy`, etc.
- Machine learning: `tensorflow` (`keras`), `scikit-learn`, `jax`
- AI assistants: Antigravity, Claude, Gemini, Deep Research, CamberCloud
- Parallelization: OpenMP, GPU (CUDA C/C++, `pycuda`), cluster computing (`slurm/launcher`)
- General: `git`, $\text{\LaTeX}$

AWARDS AND GRANTS

- ACCESS Explore allocation for the project “Reconstructing black hole shadows from sparse interferometric data” (400,000 ACCESS credits $\approx$ 400,000 core-hours)
- Space@Hopkins Seed Grant “Preparing to Untangle the Gravitational Wave Cacophony from Multivariate Data for the LISA Space Mission” (PI: E. Berti, Co-I: S. Kushnarev, Collaborators: **V. Strokov**, C. Kümmerle, N. Geissler)
- 2023 Teaching Award (Department of Physics and Astronomy, JHU)

TEACHING

- Sole Course Instructor for Advanced Electromagnetism 1 (Spring 2026) at *West Virginia University*
- Teaching assistant at *Johns Hopkins University*: graduate-level Electrodynamics (Spring 2022–2024), graduate-level Quantum Mechanics I (Fall 2022), General Physics Lab II (Spring 2021), General Physics for Physics Science Majors (Spring 2021)
- Developing and teaching an intersession course at *Johns Hopkins University*: “MCMC: a lever to move the world of data” (Intersession 2023)
- Developing and teaching physics “bootcamps” for the Post-Baccalaureate Premedical Program at *Johns Hopkins University*
- Physics tutor for the Post-Baccalaureate Premedical Program at *Johns Hopkins University* (2022–2023 academic year)
- Mentoring a high school student (Jacob Winick, Millburn High School, NJ) on a research project *Visualizing Gravity in GR* (2022)

SELECTED PRESENTATIONS

- “Hankel low-rank matrix approximation for time series denoising” (talk), *NANOGrav Fall 2025 Meeting*, Missoula, MT 2025
- “Quasimonochromatic LISA sources in the Frequency Domain” (talk), *APS April Meeting 2024*, Sacramento, CA 2024
- “LISA Constraints on an Intermediate-Mass Black Hole in the Galactic Center” (talk), *MODEST-23: Star Clusters in the Post-Pandemic Era*, Evanston, IL 2023
- “LISA Constraints on an Intermediate-Mass Black Hole in the Galactic Center” (talk), *APS April Meeting 2023*, Minneapolis, MN 2023
- “Hunting for intermediate-mass black holes with LISA binary radial velocity measurements” (talk), *Intermediate-Mass Black Holes: New Science from Stellar Evolution to Cosmology*, San Juan, Puerto Rico 2022
- “Exoplanetary candidates to observations with Millimetron” (talk), *42nd COSPAR Scientific Assembly*, Pasadena, CA 2018

LANGUAGES

Russian: Native.
English/Portuguese: Proficient.
French: Advanced.
German: Reading Knowledge.